

The large-field region in the Hamiltonian route needs an operator inequality, and the measure statement is weaker than that

Working out what the gap inequality requires from the large-field region replaces “an effective description” by something more precise, and corrects one earlier reading. Three positive facts first. **The truncation inside the large-field region costs only a longer range.** The error of truncating the flow-conjugated Hamiltonian at range Ka is $C \exp[2t\|G\|_\infty - cK]$, so on configurations with $t\|G\|_\infty \leq \eta_\infty/8$ a range $K \geq \eta_\infty/(4c) + \log(C/\delta)/c$ brings it below δ there too; the competition recorded in the transfer note §4 between $e^{-c\eta^2/g^2}$ and $e^{2\eta}$ was a fixed- K artifact, and in the window $\eta \leq C_2$ of Section 4 the range is $O(1)$. **Localizing to the small-field region is cheap per doubling.** For a smooth partition $\chi_P^2 + \chi_Q^2 = 1$ depending on the flowed block density at radius $2a$, the IMS localization formula gives $H = \chi_P H \chi_P + \chi_Q H \chi_Q - L$ with

$$L \leq C_L \frac{\hbar c}{a} \frac{g^2}{\eta^2}, \quad C_L = \frac{32\pi^2 c_0^2}{3},$$

supported on the transition region, against the $\hbar c/a$ scale of the step. **A positive perturbation that is large only on a rare set is harmless:** for $W \geq 0$, $\text{gap}(H+W) \geq \text{gap}(H) - \langle \Omega, W \Omega \rangle$, and the expectation is the measure of the support times the size. The obstruction is the sign. What the chain of lower bounds needs on the large-field region is the operator inequality

$$H - E_0 \geq c \frac{\hbar c}{a} \frac{\eta^2}{g^2} \mathbf{1}_Q - (\text{small}) \mathbf{1}_P,$$

a local energy excess, and every route to it from the Hamiltonian alone loses a term of order $(\hbar c/a) \times (\text{surface of the block})$ on the small-field region, because cutting the vacuum costs zero-point energy on every shared link. That loss is a *negative* perturbation of order $\hbar c/a$ on P , and negative perturbations of that size bind: a well of depth D on a set of ground-state measure ε captures a state once D exceeds the localization cost L , whatever ε is. So the Euclidean estimate, which is a measure statement, is strictly weaker than what the Hamiltonian chain consumes, which is an operator statement. The polymer expansion is the device that makes measure statements sufficient, by attaching the factor $e^{-c\eta^2/g^2}$ to each large-field polymer as a convergence factor rather than asking for an operator bound. That is the reason the constructive programme is Euclidean, stated as a computation. Constants explicit; nothing promoted.

1. What the gap inequality consumes

The route proves $\text{gap}(H) \geq \Delta$ by a chain of operator lower bounds: conjugation (exact), truncation ($H \geq (1-\delta)H_{\text{eff}}$ in the relative sense, since the truncated kinetic metric differs from the full one by a small multiple of it), decimation of the modes between a and $2a$ (Feshbach, Lemma 1'), and induction. The comparison with Δ is made only at the end, at strong coupling, because at any intermediate scale $\Delta = \hbar c\Lambda$ is smaller than every error the step can afford, all of which are measured against $\hbar c/a$. So each step must deliver

$$H^{(a)} \geq H_{\text{eff}}^{(2a)} - R, \quad R \text{ small against } \hbar c/a \text{ and irrelevant,}$$

and the question for the large-field region Q is what it contributes to R .

2. Localization: the IMS formula on the configuration manifold

For $H = K(-\Delta) + V$ on a Riemannian manifold and smooth $\chi_1^2 + \chi_2^2 = 1$, the IMS formula (Cycon, Froese, Kirsch and Simon, *Schrödinger Operators*, Springer 1987, Theorem 3.2; metadata level) is the identity

$$H = \chi_1 H \chi_1 + \chi_2 H \chi_2 - K(|\nabla \chi_1|^2 + |\nabla \chi_2|^2).$$

On G^ε with $K = (\hbar c/a)(g^2/2)$ per link, take $\chi_P = \cos \theta(f_B)$, $\chi_Q = \sin \theta(f_B)$ with θ rising from 0 to $\pi/2$ as the flowed block density $f_B = \ell \int_B |B_t|^2 d^3x$ rises from η^2 to $4\eta^2$, at $\ell = 2a$ and $\sqrt{8t} = \ell$. Then $|\nabla \chi_P|^2 + |\nabla \chi_Q|^2 = \theta'(f_B)^2 |\nabla f_B|^2$ with $\theta' = \pi/(6\eta^2)$, and

$$\frac{\partial f_B}{\partial \theta_{\text{link}}} = 2\ell a \int_B B_t(x) K_t(x-y) d^3x \leq 4c_0 \frac{\eta a}{\ell},$$

since a unit change of a link angle changes the flowed field by $aK_t(x-y)$ and $|B_t| \leq 2c_0\eta/\ell^2$ on the transition region. Summing over the $3(2\ell/a)^3$ links of the smearing neighbourhood, $|\nabla f_B|^2 \leq 384c_0^2\eta^2\ell/a = 768c_0^2\eta^2$ at $\ell = 2a$, so

$$L_B \leq \frac{\hbar c}{a} \frac{g^2}{2} \cdot \frac{\pi^2}{36\eta^4} \cdot 768c_0^2\eta^2 = \frac{32\pi^2c_0^2}{3} \frac{\hbar c}{a} \frac{g^2}{\eta^2}.$$

Two remarks. The general- ℓ form carries a factor ℓ/a , so localizing at a scale far above the lattice is expensive and the localization must be done one doubling at a time, in the variables of the current scale. And L_B is a multiplication operator supported on the transition region of block B , whose ground-state measure is $e^{-c\eta^2/g^2}$: it is small against $\hbar c/a$ by $C_L g^2/\eta^2$ and rare.

3. The sign of a rare perturbation

Proposition 1. If $W \geq 0$ then $\text{gap}(H+W) \geq \text{gap}(H) - \langle \Omega, W\Omega \rangle$.

Proof. $E_1(H+W) \geq E_1(H)$ by min-max, and $E_0(H+W) \leq \langle \Omega, (H+W)\Omega \rangle = E_0 + \langle \Omega, W\Omega \rangle$.

□

So a positive perturbation of size D on a set of ground-state measure ε costs the gap at most $D\varepsilon$, and with $\varepsilon = e^{-c\eta^2/g^2}$ and $D = O(\hbar c/a)$ that is negligible.

Proposition 2. A negative perturbation $-W$ with $W \geq D \mathbf{1}_A$ on a set A of ground-state measure ε produces a state of energy at most $E_0 + L_A - D$, where $L_A = K \int |\nabla \chi_A|^2 \Omega^2 / \|\chi_A \Omega\|^2$ is the localization cost of a cutoff χ_A supported in A .

Proof. The trial state $\chi_A \Omega / \|\chi_A \Omega\|$ has energy $E_0 + L_A$ in H by the exact identity $\langle \chi \Omega, (H - E_0) \chi \Omega \rangle = K \int |\nabla \chi|^2 \Omega^2$, and $-W$ lowers it by at least D . □

The measure ε does not appear in the conclusion. A negative perturbation of order $\hbar c/a$ on a rare set reorganizes the low spectrum as soon as $D > L_A$, and L_A is of order $(\hbar c/a)C_L g^2/\eta^2 \ll \hbar c/a$. **The sign of the error on the large-field region decides everything, and the measure of the region decides nothing.**

4. The operator inequality the chain needs, and its cost

For the large-field sector to contribute only harmlessly to R , the step needs

$$\chi_Q(H - E_0)\chi_Q \geq c_1 \frac{\hbar c}{a} \frac{\eta^2}{g^2} \chi_Q^2,$$

a *local energy excess*: configurations with a large flowed field in a block have energy above the vacuum by the field's energy. This is physically evident at weak coupling and is the Hamiltonian form of the action lower bound of the lower-bound note. Its proof from the Hamiltonian splits $H = H_N + H'$ with N the neighbourhood of the block and uses two facts: $V_N \geq (\hbar c/a)(2/g^2)c_1\eta^2$ on Q , by three-dimensional flow monotonicity and positivity, exactly as in the Euclidean case; and an upper bound on $E_0 - E_0(H - V_N)$, the energy the neighbourhood's magnetic terms add to the vacuum. The second is where the loss occurs. The crude bound uses the ground state of $H - V_N$, which is Haar-flat on the interior links and gives $\langle V_N \rangle_{\text{Haar}} = (\hbar c/a)(2N/g^2)|N|_p$, of the same order $1/g^2$ as the excess and useless. The correct order is the zero-point energy, $O(1) \cdot (\hbar c/a)|N|_p$, and any trial state that achieves it must extend the outside vacuum smoothly into N ; the cut then costs $O(\hbar c/a)$ per shared link, an amount

$$C_V \frac{\hbar c}{a} |\partial N|$$

that is subtracted *everywhere*, including on P . The resulting inequality is

$$H - E_0 \geq \left[\frac{2c_1\eta^2}{g^2} - C_V|N|_p \right] \frac{\hbar c}{a} \mathbf{1}_Q - C_V|\partial N| \frac{\hbar c}{a} \mathbf{1}_P,$$

and the second term is a negative perturbation of order $\hbar c/a$ on the small-field region, of exactly the kind Proposition 2 says cannot be tolerated. The window in which the first term is positive,

$$\eta \geq C_1 g, \quad C_1^2 = \frac{C_V|N|_p}{2c_1} \simeq 160 \text{ for a } 4a \text{ neighbourhood,}$$

together with the small-angle requirement of the perturbative decimation on P , $\eta/4 \leq C_2$, is nonempty only for $g \leq 4C_2/C_1$, that is $1/g^2 \gtrsim 10^2$ with these crude constants. Even inside the window the P -side loss remains.

5. Why the Euclidean framework escapes this

The Euclidean side proves a measure statement, $\mathbb{P}(Q) \leq e^{-c\eta^2/g^2}$, and never an operator one. It can afford this because the cluster (polymer) expansion consumes measure statements directly: a large-field polymer enters the expansion with the weight $e^{-c\eta^2/g^2}$ as a convergence factor, and the effective action outside the polymers is computed perturbatively. No inequality of the form $H - E_0 \geq (\dots)\mathbf{1}_Q$ is ever needed, because there is no Hamiltonian to bound; the gap is read off at the end from the decay of the two-point function, which is again a measure statement.

The Hamiltonian chain of Section 1, by contrast, consumes operator inequalities, and Section 3 shows it cannot substitute a measure statement for them when the sign is wrong. This is the precise content of “the constructive programme is Euclidean”: **the large-field region is controllable in measure and uncontrollable in norm, and the polymer expansion is the device that makes measure control sufficient.**

6. Consequence for STATE

The obligation “an effective description inside the large-field region” is replaced by the following. The truncation there is cheap (range $O(\eta)$), the localization is cheap ($C_L g^2/\eta^2$ per doubling), and positive rare errors are harmless. The chain needs the local energy-excess operator inequality, whose Hamiltonian proof loses $C_V|\partial N|\hbar c/a$ on the small-field region, a negative perturbation of the order that binds. So the Hamiltonian route, as a chain of operator lower bounds, cannot close through the large-field region, and the Euclidean route closes it by consuming measure statements in a polymer expansion. The honest division of labour is: Hamiltonian for T1, T2, the small-volume theorem, the upper bounds and the final gap extraction; Euclidean polymer expansion for the renormalization steps. The coupling map is then: weak-coupling expansion for $1/g^2 \gtrsim C_1^2/(16C_2^2)$, strong-coupling theorem for $54/g^4 \leq \beta_*$, and the intermediate region between them, in which the gap forms and no expansion applies; its width in one-loop doublings is about 10^3 with the crude constants above and about 20 if the weak side reaches $g^2 \sim 1/2$ (threshold note §4).